1) Develop a program to Load a dataset and select one numerical column. Compute mean, median, mode, standard deviation, variance, and range for a given numerical column in a dataset. Generate a histogram and boxplot to understand the distribution of the data. Identify any outliers in the data using IQR. Select a categorical variable from a dataset. Compute the frequency of each category and display it as a bar chart or pie chart.

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Create the dataset based on the provided data
data = pd.DataFrame({
    'Age': [23, 23, 27, 27, 39, 41, 47, 49, 50, 52, 54, 54, 56, 57, 58, 58, 60, 61],
    '%Fat': [7.8, 9.5, 17.8, 25.9, 26.5, 27.2, 27.4, 28.8, 30.2, 31.2, 31.4, 32.9, 33.4, 34.1,
34.6, 35.7, 41.2, 42.5]
})

# Save the dataset to a CSV file
data.to_csv('Age_Fat.csv', index=False)

# Load the dataset
print("Dataset Loaded from Age_Fat.csv")
data = pd.read_csv('Age_Fat.csv')

# Preview the dataset
print("Dataset Preview:")
print(data.head())

# Numerical column: %Fat
numerical_column = '%Fat'
```

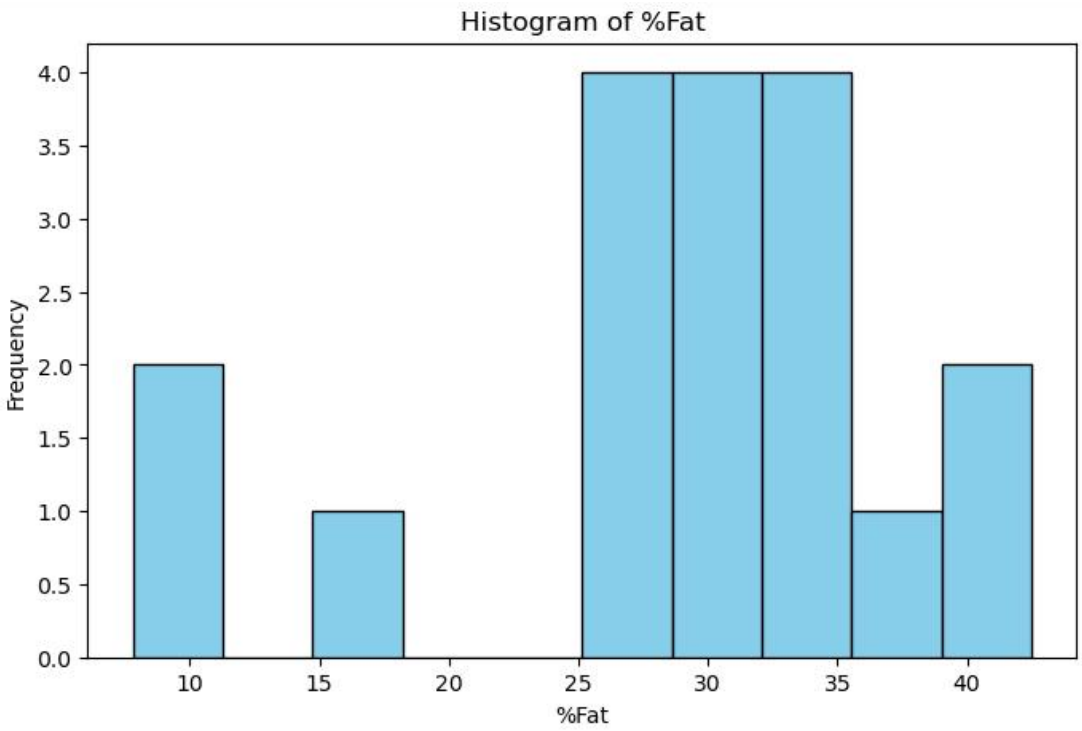
```
data_num = data[numerical_column]

# Compute statistics
mean_val = data_num.mean()
median_val = data_num.median()
mode_val = data_num.mode()[0]
std_dev = data_num.std()
variance = data_num.var()
range_val = data_num.max() - data_num.min()

# Print statistics
print("\nStatistics for Numerical Column:")
print(f"Mean: {mean_val}")
print(f"Median: {median_val}")
print(f"Mode: {mode_val}")
print(f"Standard Deviation: {std_dev}")
print(f"Variance: {variance}")
print(f"Range: {range_val}")

# Generate a histogram
plt.figure(figsize=(8, 5))
plt.hist(data_num, bins=10, color='skyblue', edgecolor='black')
plt.title(f"Histogram of {numerical_column}")
plt.xlabel(numerical_column)
plt.ylabel("Frequency")
plt.show()

# Generate a boxplot
plt.figure(figsize=(8, 5))
sns.boxplot(x=data_num, color='lightgreen')
plt.title(f"Boxplot of {numerical_column}")
plt.show()

# Identify outliers using IQR
```

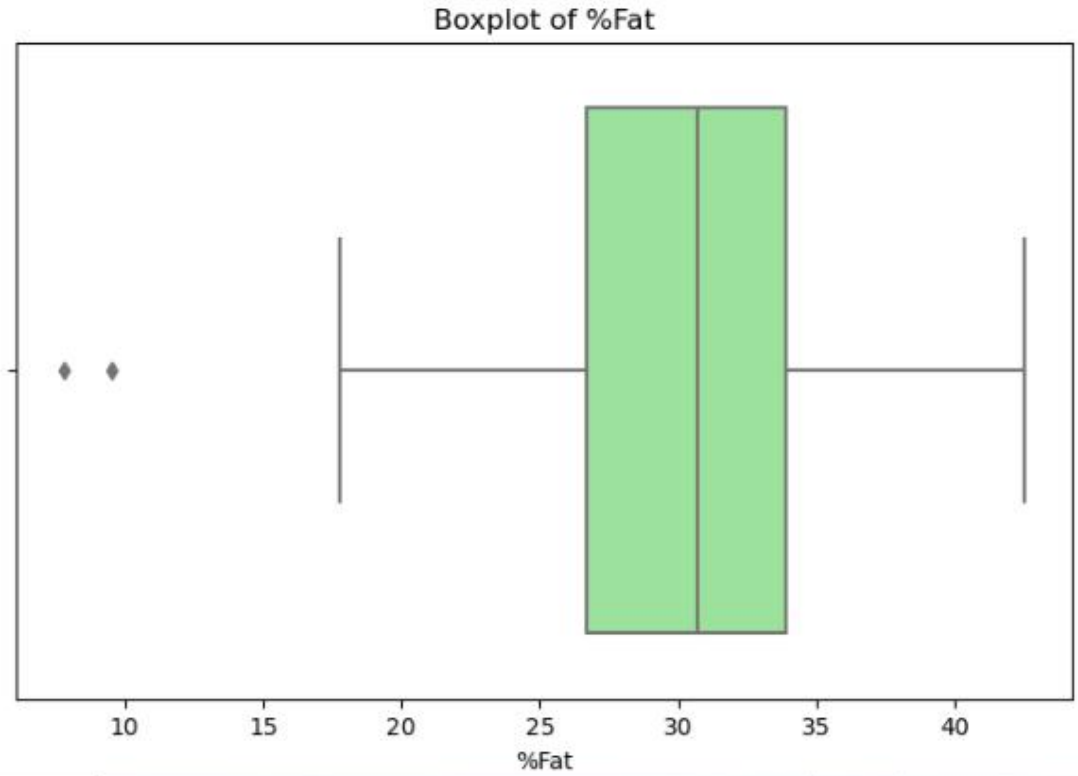
```
q1 = data_num.quantile(0.25)
q3 = data_num.quantile(0.75)
iqr = q3 - q1
lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr
outliers = data_num[(data_num < lower_bound) | (data_num > upper_bound)]

print("\nOutliers:")
print(outliers)

# Add a categorical column (Age Group) for demonstration purposes
def age_group(age):
    if age < 30:
        return 'Young'
    elif 30 <= age <= 50:
        return 'Middle-aged'
    else:
        return 'Older'

data['Age Group'] = data['Age'].apply(age_group)

# Categorical column: Age Group
categorical_column = 'Age Group'
data_cat = data[categorical_column]

# Compute frequency of each category
category_counts = data_cat.value_counts()

print("\nCategory Frequencies:")
print(category_counts)

# Bar chart for the categorical column
plt.figure(figsize=(8, 5))
category_counts.plot(kind='bar', color='coral', edgecolor='black')
```

```
plt.title(f"Bar Chart of {categorical_column}")
plt.xlabel(categorical_column)
plt.ylabel("Frequency")
plt.show()

# Pie chart for the categorical column
plt.figure(figsize=(8, 5))
category_counts.plot(kind='pie', autopct='%1.1f%%', startangle=90,
colors=sns.color_palette('pastel'))
plt.title(f"Pie Chart of {categorical_column}")
plt.ylabel("")  # Remove y-axis label
plt.show()
```

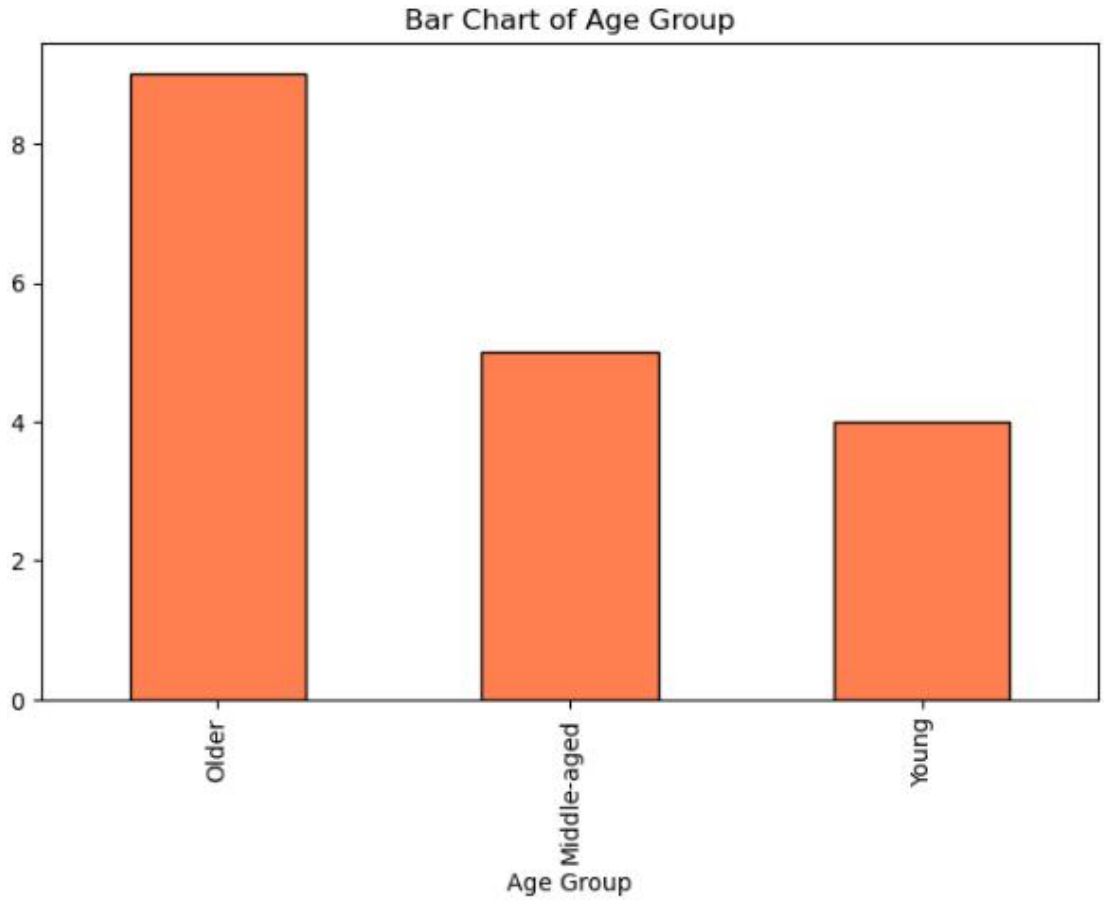
Output:

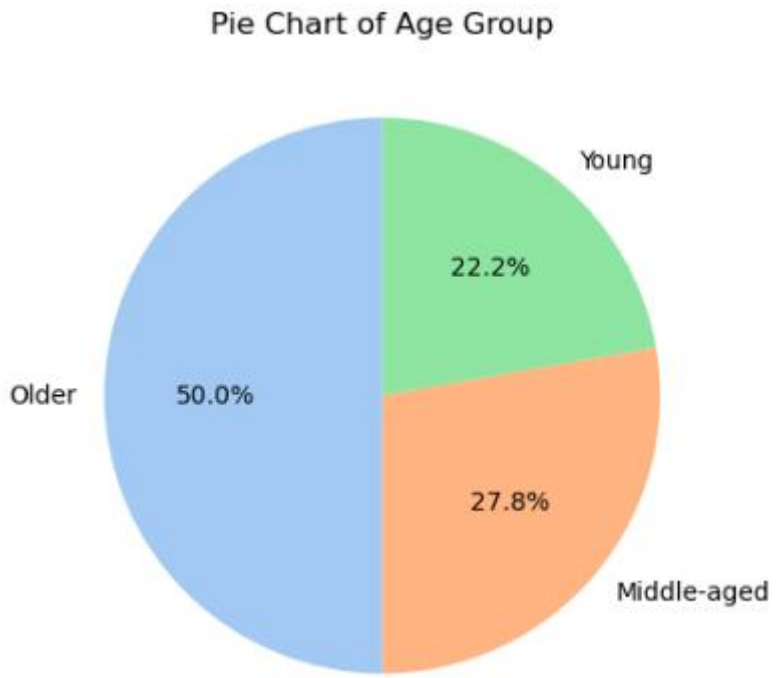
```
Dataset Loaded from Age_Fat.csv
Dataset Preview:
   Age  %Fat
0   23   7.8
1   23   9.5
2   27  17.8
3   27  25.9
4   39  26.5

Statistics for Numerical Column:
Mean: 28.783333333333328
Median: 30.7
Mode: 7.8
Standard Deviation: 9.2543948224296
Variance: 85.64382352941178
Range: 34.7
```


Histogram of %Fat
Frequency
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
10
15
20
25
30
35
40
%Fat


Boxplot of %Fat
10
15
20
25
30
35
40
%Fat

```
Outliers:
0    7.8
1    9.5
Name: %Fat, dtype: float64

Category Frequencies:
Age Group
Older          9
Middle-aged    5
Young          4
Name: count, dtype: int64
```





jupyter Age_Fat.csv

File Edit View Language

```
Age,%Fat
23,7.8
23,9.5
27,17.8
27,25.9
39,26.5
41,27.2
47,27.4
49,28.8
50,30.2
52,31.2
54,31.4
54,32.9
56,33.4
57,34.1
58,34.6
58,35.7
60,41.2
61,42.5
```

2) Develop a program to load the Iris dataset. Implement the k-Nearest Neighbors (k-NN) algorithm for classifying flowers based on their features. Split the dataset into training and testing sets and evaluate the model using metrics like accuracy and F1-score. Test it for different values of *k* (e.g., k=1,3,5) and evaluate the accuracy. Extend the k-NN algorithm to assign weights based on the

distance of neighbors (e.g., *weight* = $1/d^2$). Compare the performance of weighted k-NN and regular k-NN on a synthetic or real-world dataset.

Code:

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, f1_score, confusion_matrix
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import StandardScaler, LabelEncoder

# Load the Iris dataset from a CSV file
file_path = 'IRIS.csv'
df = pd.read_csv(file_path)

# Separate features (X) and labels (y)
X = df.iloc[:, :-1].values
y = df.iloc[:, -1].values

# Convert species labels to numerical values
label_encoder = LabelEncoder()
y = label_encoder.fit_transform(y)

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
# Standardize features
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# --------- Custom 1/d^2 Weight Function ----------
def inverse_square_weight(distances):
    # Adding small epsilon to avoid division by zero
```

```
    return 1 / (distances**2 + 1e-5)
# ------------------------------------------------
# Function to evaluate k-NN
def evaluate_knn(X_train, X_test, y_train, y_test, k_values, weighted=False):
    results = {}

    for k in k_values:

        if weighted:
            knn = KNeighborsClassifier(
                n_neighbors=k,
                weights=inverse_square_weight   # <-- 1/d^2 applied here
            )
        else:
            knn = KNeighborsClassifier(n_neighbors=k)

        # Train model
        knn.fit(X_train, y_train)

        # Predict
        y_pred = knn.predict(X_test)

        # Evaluate
        accuracy = accuracy_score(y_test, y_pred)
        f1 = f1_score(y_test, y_pred, average='weighted')
        conf_matrix = confusion_matrix(y_test, y_pred)

        results[k] = {
            'accuracy': accuracy,
            'f1_score': f1,
            'conf_matrix': conf_matrix
        }

    return results
```

```
# Test different k values
k_values = [1, 3, 5]

# Regular k-NN
regular_knn_results = evaluate_knn(
    X_train, X_test, y_train, y_test, k_values, weighted=False
)

print("Regular k-NN Results:")
for k, metrics in regular_knn_results.items():
    print(f"\nk={k}: Accuracy={metrics['accuracy']:.4f}, "
          f"F1-Score={metrics['f1_score']:.4f}")
    print("Confusion Matrix:")
    print(metrics['conf_matrix'])

# Weighted k-NN (1/d^2)
weighted_knn_results = evaluate_knn(
    X_train, X_test, y_train, y_test, k_values, weighted=True
)

print("\nWeighted k-NN Results (1/d^2):")
for k, metrics in weighted_knn_results.items():
    print(f"\nk={k}: Accuracy={metrics['accuracy']:.4f}, "
          f"F1-Score={metrics['f1_score']:.4f}")
    print("Confusion Matrix:")
    print(metrics['conf_matrix'])
```

Output:

```
Regular k-NN Results:
k=1: Accuracy=0.9778, F1-Score=0.9777
Confusion Matrix:
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]
```

k=3: Accuracy=1.0000, F1-Score=1.0000

Confusion Matrix:

```
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

k=5: Accuracy=1.0000, F1-Score=1.0000

Confusion Matrix:

```
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

Weighted k-NN Results (1/d^2):

k=1: Accuracy=0.9778, F1-Score=0.9777

Confusion Matrix:

```
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]
```

k=3: Accuracy=1.0000, F1-Score=1.0000

Confusion Matrix:

```
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

k=5: Accuracy=1.0000, F1-Score=1.0000

Confusion Matrix:

```
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

3) Develop a program to Load a dataset with at least two numerical columns (e.g., Iris, Titanic). Plot a scatter plot of two variables and calculate their Pearson cor

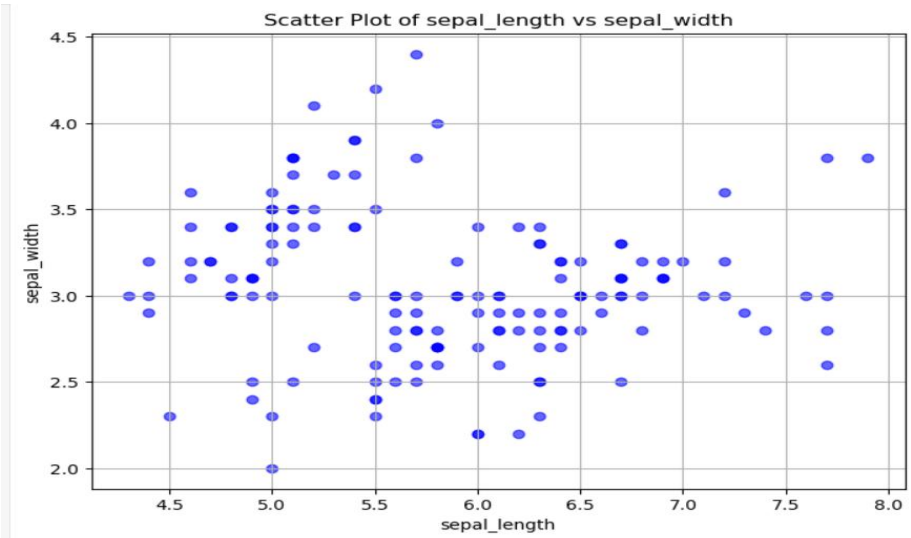
relation coefficient. Write a program to compute the covariance and correlation matrix for a dataset. Visualize the correlation matrix using a heatmap to know w hich variables have strong positive/negative correlations.

Code:

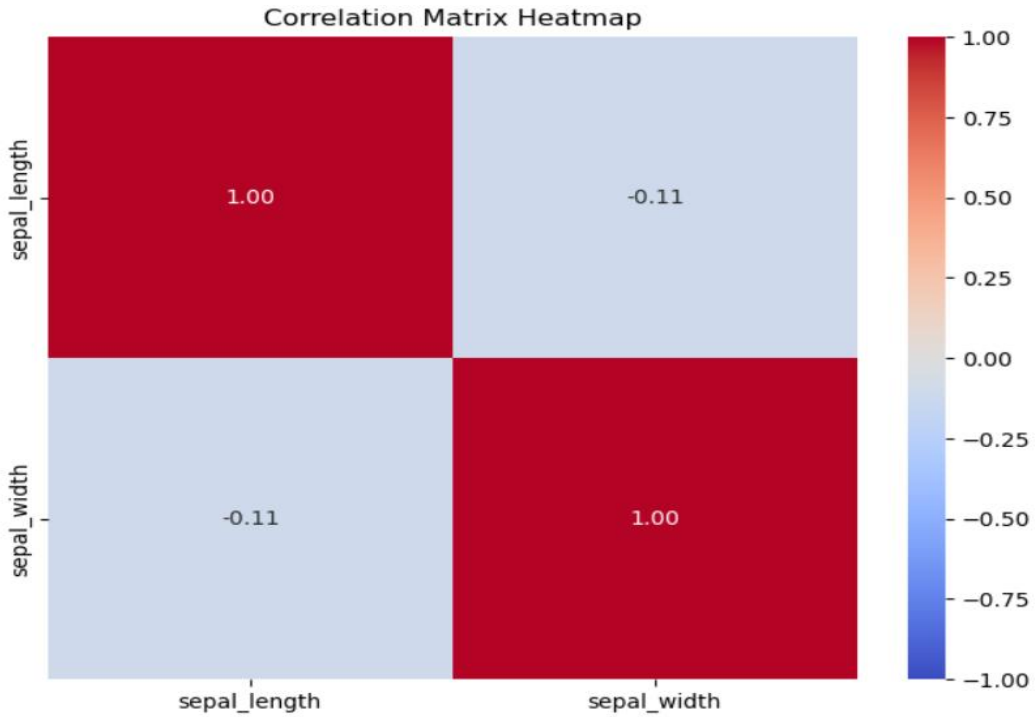
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset (update 'your_dataset.csv' with your actual file path)
#file_path = "Sample_Marks.csv"  # Replace with your dataset file path
file_path = "IRIS.csv"
df = pd.read_csv(file_path)

# Extract two columns (update 'Column1' and 'Column2' with actual column names)
#column1 = "Test Score"  # Replace with the first column name
#column2 = "Video Game"  # Replace with the second column name
column1 = "sepal_length"
column2 = "sepal_width"
selected_data = df[[column1, column2]]

# Scatter plot of the selected columns
plt.figure(figsize=(8, 6))
plt.scatter(selected_data[column1], selected_data[column2], color="blue", alpha=0.6)
#plt.scatter(selected_data[column1], selected_data[column2], color="blue")
plt.title(f"Scatter Plot of {column1} vs {column2}")
plt.xlabel(column1)
plt.ylabel(column2)
plt.grid(True)
plt.show()

# Calculate Pearson correlation coefficient
correlation_coefficient = np.corrcoef(selected_data[column1], selected_data[column
2])[0, 1]
```

```
print(f"Pearson Correlation Coefficient ({column1} vs {column2}): {correlation_coef
ficient}")

# Compute covariance matrix
cov_matrix = np.cov(selected_data[column1], selected_data[column2])
print("\nCovariance Matrix:")
print(cov_matrix)

# Compute correlation matrix
correlation_matrix = selected_data.corr()
print("\nCorrelation Matrix:")
print(correlation_matrix)

# Visualize the correlation matrix using a heatmap
plt.figure(figsize=(8, 6))
#sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm", fmt=".2f", cbar=Tr
ue)
#sns.heatmap(correlation_matrix, annot=True, cmap="viridis", fmt=".2f", cbar=True)
sns.heatmap(correlation_matrix,cmap="coolwarm",vmin=-1,vmax=1, annot=True, fm
t=".2f", cbar=True)
#sns.heatmap(correlation_matrix,cmap="RdBu",vmin=-1,vmax=1, annot=True)
plt.title("Correlation Matrix Heatmap")
plt.show()
```

Output:



```
Pearson Correlation Coefficient (sepal_length vs sepal_width): -0.10936924995064937

Covariance Matrix:
[[ 0.68569351 -0.03926846]
 [-0.03926846  0.18800403]]

Correlation Matrix:
              sepal_length  sepal_width
sepal_length      1.000000    -0.109369
sepal_width      -0.109369     1.000000
```



4) Develop a program to implement Principal Component Analysis (PCA) for reducing the dimensionality of the Iris dataset from 4 features to 2.

Code:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

# Load the dataset from CSV file
df = pd.read_csv('iris.csv')  # Replace 'dataset.csv' with your actual file name
```

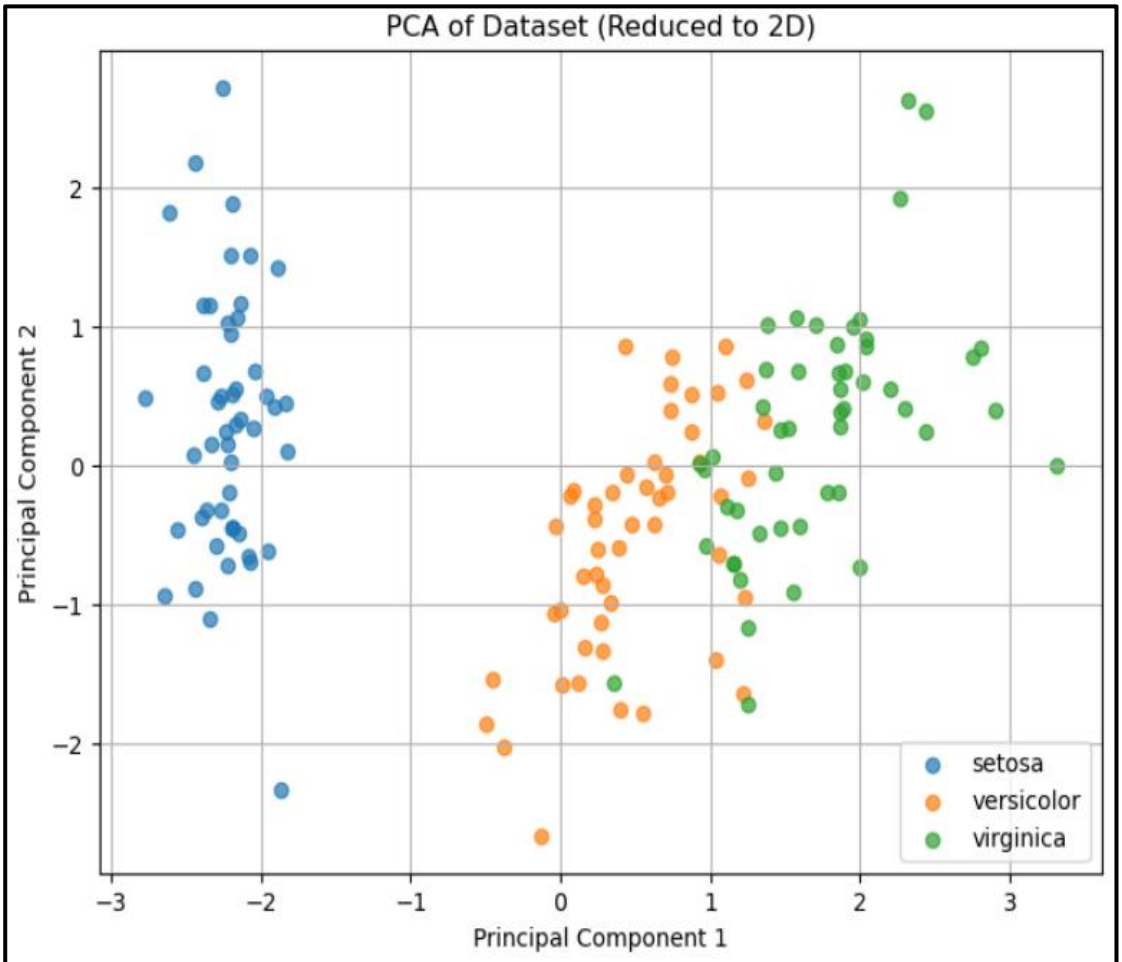
```
# Display the first few rows of the dataset to understand its structure
print("Dataset preview:")
print(df.head())

# Assuming the last column is the target variable
X = df.iloc[:, :-1].values  # Features
y = df.iloc[:, -1].values  # Labels

# Standardize the data (mean=0, variance=1)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Apply PCA to reduce dimensionality to 2
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

# Get unique labels for coloring
unique_labels = np.unique(y)

# Plot the PCA-reduced data
plt.figure(figsize=(8, 6))
for target in unique_labels:
    plt.scatter(X_pca[y == target, 0], X_pca[y == target, 1], label=target, alpha=0.7)

# Print the shape of the original and reduced datasets
print("\n\nOriginal dataset shape:", X_scaled.shape)  # Before PCA
print("Reduced dataset shape:", X_pca.shape)  # After PCA

plt.title('PCA of Dataset (Reduced to 2D)')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.legend()
plt.grid(True)
plt.show()
```

Print the explained variance ratio

print("Explained variance ratio:", pca.explained_variance_ratio_)

Output:

```
Dataset preview:
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa


Original dataset shape: (150, 4)
Reduced dataset shape: (150, 2)
```




```
Explained variance ratio: [0.72770452 0.23030523]
```

5) Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs.

Algorithm:

1. Read the Given data Sample to X and the curve (linear or non linear) to Y

2. Set the value for Smoothening parameter or Free parameter say τ

3. Set the bias /Point of interest set x0 which is a subset of X

4. Determine the weight matrix using :

$$w(x, x_o) = e^{-\frac{(x-x_o)^2}{2\tau^2}}$$

5. Determine the value of model term parameter β using:

$$\beta(x_o) = (X^T W X)^{-1} X^T W y$$

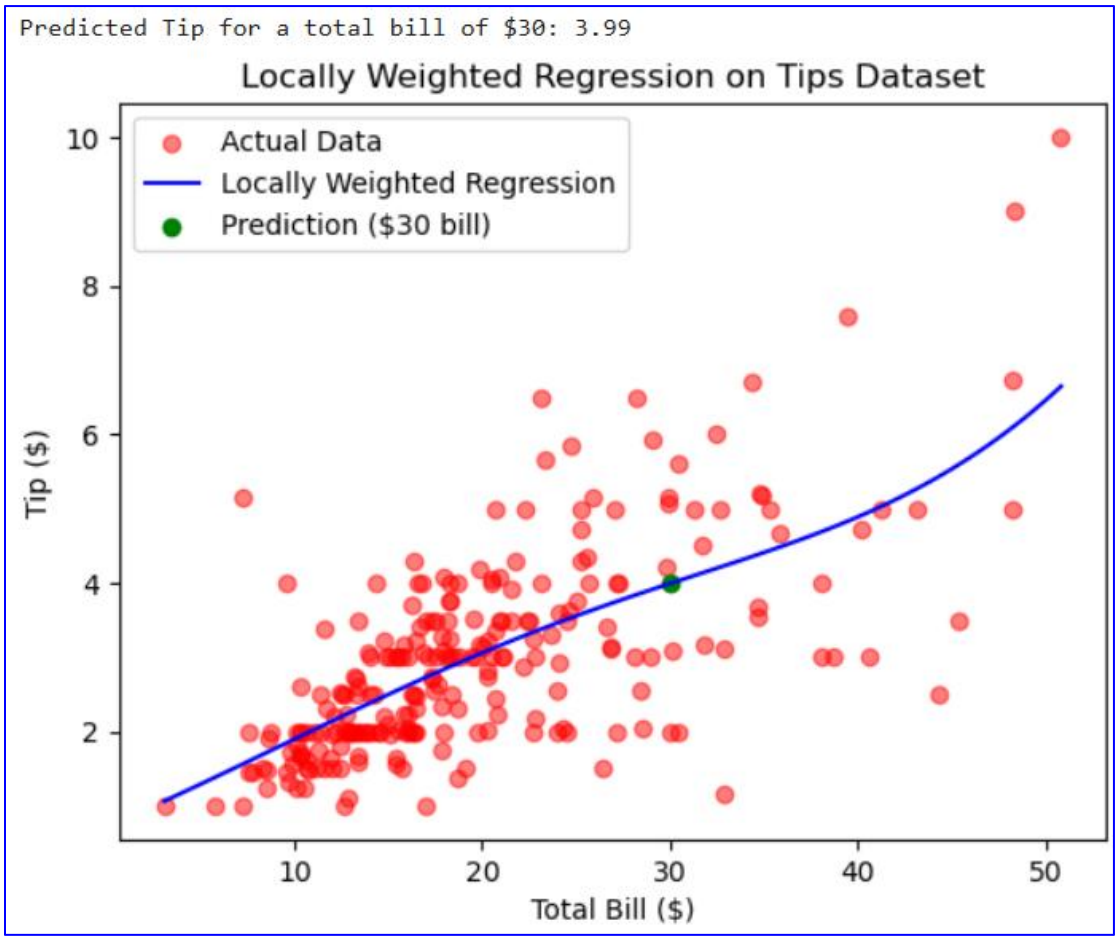
6. Prediction = x_0*β

Code:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Load the dataset from CSV file
file_path = "10-dataset.csv"  # Update this with the actual path to your CSV file
tips = pd.read_csv(file_path)

# Use total_bill as feature (X) and tip as target (y)
X = tips["total_bill"].values
y = tips["tip"].values

# Locally Weighted Regression Function
def locally_weighted_regression(x_query, X, y, tau):
    m = len(X)
```

```
    weights = np.exp(-((X - x_query) ** 2) / (2 * tau ** 2))  # Compute weights

    # Create weighted linear regression
    X_b = np.c_[np.ones(m), X]  # Add bias term (column of ones)
    W = np.diag(weights)  # Create diagonal weight matrix

    # Solve for theta using normal equation: theta = (X^T W X)^-1 X^T W y
    #theta = np.linalg.inv(X_b.T @ W @ X_b) @ (X_b.T @ W @ y)
    theta = np.linalg.pinv(X_b.T @ W @ X_b) @ (X_b.T @ W @ y)

    # Predict tip for x_query
    x_query_b = np.array([1, x_query])
    return x_query_b @ theta

# Predicting tip for a total bill of $30
tau = 10  # Bandwidth (adjust as needed)
x_query = 30
predicted_tip = locally_weighted_regression(x_query, X, y, tau)
print(f"Predicted Tip for a total bill of $30: {predicted_tip:.2f}")

# Plot the results
X_range = np.linspace(X.min(), X.max(), 100)
y_pred = np.array([locally_weighted_regression(x, X, y, tau) for x in X_range])

plt.scatter(X, y, color='red', alpha=0.5, label="Actual Data")
plt.plot(X_range, y_pred, color='blue', label="Locally Weighted Regression")
plt.scatter([x_query], [predicted_tip], color='green', marker='o', label="Prediction
($30 bill)")
plt.xlabel("Total Bill ($)")
plt.ylabel("Tip ($)")
plt.legend()
plt.title("Locally Weighted Regression on Tips Dataset")
plt.show()
```

Output:



6) Develop a program to demonstrate the working of Linear Regression and Polynomial Regression. Use Boston Housing Dataset for Linear Regression and Auto MPG Dataset (for vehicle fuel efficiency prediction) for Polynomial Regression.

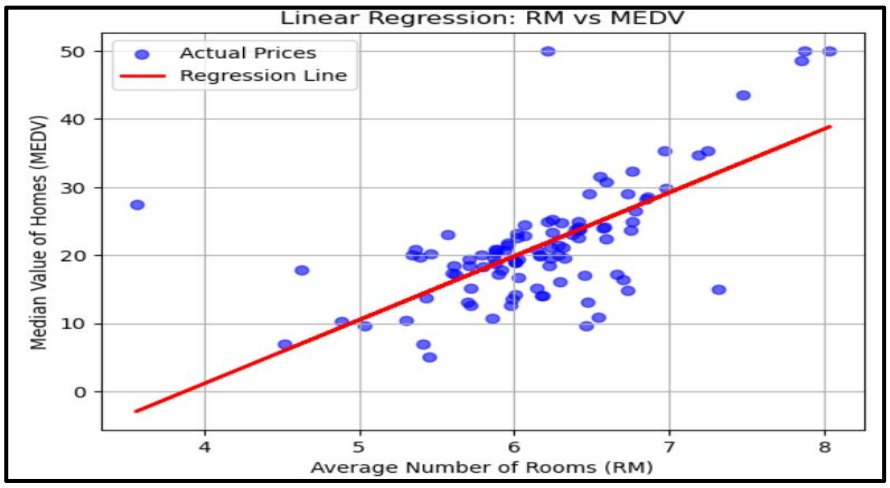
(i) Linear Regression:

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

# Load dataset
data = pd.read_csv("BostonHousing.csv")
```

```
# Select one feature and target
X = data[['rm']]      # Average number of rooms
y = data['medv']      # Median home value

# Split into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Plot
plt.scatter(X_test, y_test, color='blue', label='Actual Prices', alpha=0.6)
plt.plot(X_test, y_pred, color='red', linewidth=2, label='Regression Line')
plt.xlabel("Average Number of Rooms (RM)")
plt.ylabel("Median Value of Homes (MEDV)")
plt.title("Linear Regression: RM vs MEDV")
plt.legend()
plt.grid(True)
plt.show()
```

Output:



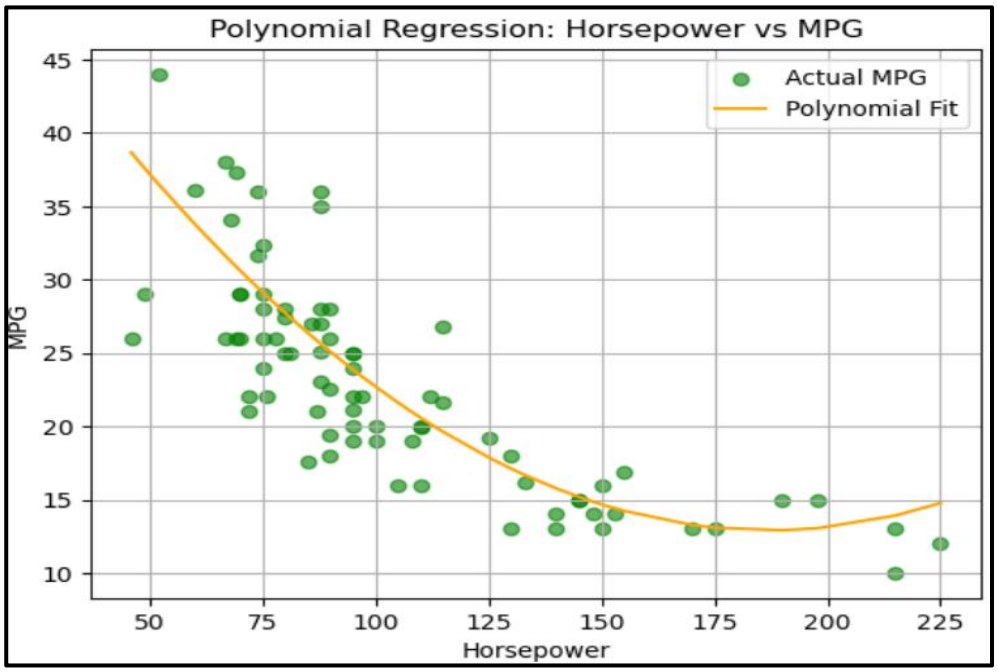
(ii) Polynomial Regression:

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.model_selection import train_test_split

# Load the dataset
data = pd.read_csv("auto-mpg.csv")

# Clean the 'horsepower' column
data.replace({'horsepower': {'?': np.nan}}, inplace=True)
data.dropna(subset=['horsepower'], inplace=True)
data['horsepower'] = data['horsepower'].astype(float)

# Select features and target
X = data[['horsepower']]
y = data['mpg']

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

# Polynomial transformation (degree 2)
poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)

# Train model
model = LinearRegression()
model.fit(X_train_poly, y_train)
```

```
# Predict
y_pred = model.predict(X_test_poly)

# Plot
plt.scatter(X_test, y_test, color='green', label='Actual MPG', alpha=0.6)
sorted_idx = X_test['horsepower'].argsort()
plt.plot(X_test.iloc[sorted_idx], y_pred[sorted_idx], color='orange', label='Polynomial
Fit')
plt.xlabel("Horsepower")
plt.ylabel("MPG")
plt.title("Polynomial Regression: Horsepower vs MPG")
plt.legend()
plt.grid(True)
plt.show()
```

Output:



7) Develop a program to load the Titanic dataset. Split the data into training and test sets. Train a decision tree classifier. Visualize the tree structure. Evaluate accuracy, precision, recall, and F1-score.

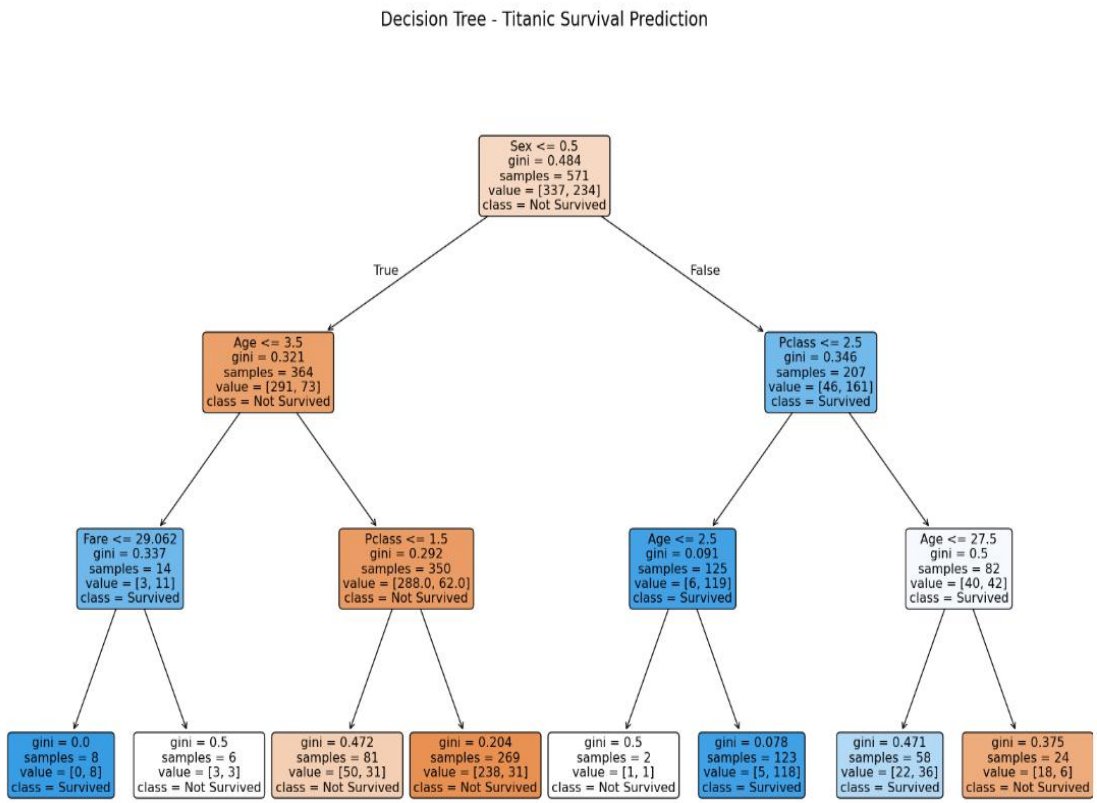
Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
```

```
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
import matplotlib.pyplot as plt

# Step 1: Load dataset
file_path = 'titanic.csv'
df = pd.read_csv(file_path)

# Step 2: Preprocess data (cleaning and encoding)
# Keep only required columns
df = df[['Survived', 'Pclass', 'Sex', 'Age', 'Fare']]

# Drop rows with any missing values in the selected columns
df.dropna(subset=['Survived', 'Pclass', 'Sex', 'Age', 'Fare'], inplace=True)

# Convert 'Sex' to numeric
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})

# Define features and label
X = df[['Pclass', 'Sex', 'Age', 'Fare']]
y = df['Survived']

# Step 3: Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=4
2)

# Step 4: Train the Decision Tree Classifier
#clf = DecisionTreeClassifier(random_state=42)
clf = DecisionTreeClassifier(random_state=42, max_depth=3)
clf.fit(X_train, y_train)

# Step 5: Visualize the decision tree
plt.figure(figsize=(20, 12))  #(Width, Height), Larger figure for clarity
plot_tree(clf,
```

```
        feature_names=X.columns,
        class_names=["Not Survived", "Survived"],
        filled=True,
        rounded=True,
        fontsize=12)
plt.title("Decision Tree - Titanic Survival Prediction\n\n", fontsize=16)
plt.show()

# Step 6: Evaluate the model
y_pred = clf.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("F1-score:", f1_score(y_test, y_pred))
```

Output:




```
Accuracy: 0.7412587412587412
Precision: 0.6938775510204082
Recall: 0.6071428571428571
F1-score: 0.6476190476190476
```

8) Develop a program to implement the Naive Bayesian classifier considering Iris dataset for training. Compute the accuracy of the classifier, considering the test data.

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score

# Step 1: Read the CSV file
data = pd.read_csv('iris.csv')

# Step 2: Separate features and target
# Assuming the last column is the target (species)
X = data.iloc[:, :-1]  # all rows, all columns except the last
y = data.iloc[:, -1]   # all rows, last column

# Step 3: Split into training and testing data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
random_state=42)

# Step 4: Train the Naive Bayes classifier
model = GaussianNB()
model.fit(X_train, y_train)

# Step 5: Make predictions
y_pred = model.predict(X_test)

# Step 6: Compute the accuracy
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy of the Naive Bayes Classifier: {accuracy * 100:.2f}%")
```

Output:

Accuracy of the Naive Bayes Classifier: 97.78%

9) Develop a program to implement k-means clustering using Wisconsin Breast Cancer data set and visualize the clustering result.

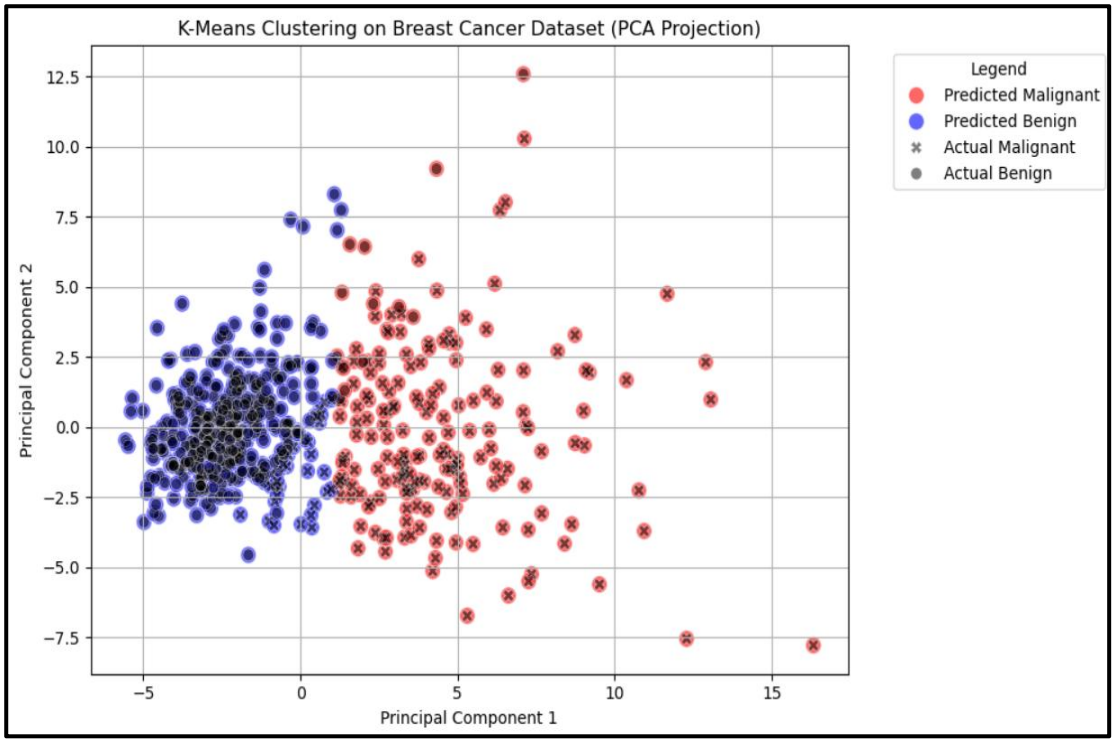
Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.metrics import accuracy_score
from scipy.stats import mode

# === Step 1: Read CSV file from device ===
df = pd.read_csv("Breast Cancer Wisconsin.csv")

# === Step 2: Preprocess the dataset ===
df = df.drop(columns=['id', 'Unnamed: 32'], errors='ignore')

if 'diagnosis' in df.columns:
    diagnosis = df['diagnosis'].map({'M': 1, 'B': 0})  # 1: Malignant, 0: Benign
    df = df.drop(columns=['diagnosis'])
else:
    diagnosis = None

# === Step 3: Standardize features ===
scaler = StandardScaler()
scaled_data = scaler.fit_transform(df)

# === Step 4: Apply K-Means Clustering ===
kmeans = KMeans(n_clusters=2, random_state=42)
kmeans.fit(scaled_data)
labels = kmeans.labels_
```

```
# === Step 5: PCA for Visualization ===
pca = PCA(n_components=2)
pca_data = pca.fit_transform(scaled_data)

# === Step 5.1: Prepare data for plotting ===
vis_df = pd.DataFrame({
    'PCA1': pca_data[:, 0],
    'PCA2': pca_data[:, 1],
    'Cluster': labels
})

if diagnosis is not None:
    vis_df['Actual'] = diagnosis

    # Map cluster numbers to label names based on majority
    cluster_map = {}
    for cluster in [0, 1]:
        majority = mode(diagnosis[labels == cluster], keepdims=True).mode[0]
        cluster_map[cluster] = 'Predicted Malignant' if majority == 1 else 'Predicted
Benign'
    vis_df['Cluster_Label'] = vis_df['Cluster'].map(cluster_map)

    # Map actual labels to strings
    vis_df['Actual_Label'] = vis_df['Actual'].map({1: 'Actual Malignant', 0: 'Actual
Benign'})

# === Step 6: Plotting with Improved Legend ===
plt.figure(figsize=(10, 6))

# First plot predicted cluster (colored)
sns.scatterplot(
    data=vis_df,
    x='PCA1', y='PCA2',
```

```
    hue='Cluster_Label',
    palette={'Predicted Malignant': 'red', 'Predicted Benign': 'blue'},
    s=100,
    alpha=0.6,
    legend='full'
)

# Then plot actual values (shapes in black)
sns.scatterplot(
    data=vis_df,
    x='PCA1', y='PCA2',
    style='Actual_Label',
    markers={'Actual Malignant': 'X', 'Actual Benign': 'o'},
    color='black',
    s=50,
    alpha=0.5,
    legend='brief'
)

plt.title("K-Means Clustering on Breast Cancer Dataset (PCA Projection)")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend(title="Legend", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.grid(True)
plt.tight_layout()
plt.show()
```

Output:



OPEN ENDED EXPERIMENTS

NOTE: Students are required to complete **one open-ended experiment** from the two listed questions. Please read the instructions carefully Before proceeding.

1) You are given a dataset such as the Titanic Dataset / Iris Dataset.

Tasks:

(a) Perform necessary data preprocessing (handling missing values, encoding, etc.).

(b) Apply at least three different machine learning algorithms, such as:

- Decision Tree
- Logistic Regression
- K-Nearest Neighbors (KNN)

(c) Evaluate each model using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score

(d) Compare the performance of all models.

Note:

- You are free to use any additional models or techniques beyond those listed.
- Analyze and justify:

- Which model performed the best?
- Why did it perform better compared to others?
- What factors (data characteristics, overfitting, etc.) influenced the results?

2) You are provided with a dataset containing raw features.

Tasks:

(a) Perform basic preprocessing of the dataset.

(b) Create at least 3–5 new features from the existing data. Examples include:

- Converting continuous variables into categories (e.g., Age → Age Group)
- Combining multiple features
- Creating meaningful derived attributes

(c) Train a machine learning model before and after feature engineering.

(d) Compare the performance of the model using metrics such as:

- Accuracy
- Precision
- Recall
- F1-score

Note:

- You are free to decide:
  - Which features to create
  - How to transform the data
- There is no single correct answer—focus on experimentation and reasoning.
- Justify:
  - Why you created those features
  - How they impacted model performance
- You can use: Titanic Dataset/House Prices Dataset/Adult Income Dataset/Student Performance Dataset/Mall Customers Dataset